Ambuj Kumar

AI Systems Architect · Generative AI / LLM Engineer · Technology Consultant

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SUMMARY

AI systems engineer and B.Tech (Electronics & Communication) student at LNMIIT who designs and builds complete enterprise software independently, from data model and backend to interface and deployment, and runs it in production. Works the way a consultant does: starts from the business problem, understands what it costs today, then builds the system that removes it, usually replacing a licensed product with one the company owns. Fine-tunes and deploys LLMs, designs data platforms, and has delivered systems in supply chain, IT operations, finance, retail and security.

EXPERIENCE

AI & Systems Engineer

Jun 2026 – 31 Aug 2026

D'Decor Home Fabrics Pvt. Ltd., Mumbai, India (India's largest home-furnishing fabric manufacturer)

- **Demand Forecasting:** Planning for around 21,000 SKUs relied on judgement and prior-year sales. Built LightGBM quantile models that give planners a probable demand range per SKU instead of a single estimate, with bias tracking.
- **Reorder Automation:** Replenishment was triggered manually on low stock. Built an optimizer that converts forecasts into purchase and production orders within safety stock, MOQ clubbing, greige/beam routing and dye-lot rules.
- **ITSM:** Hundreds of Windows and Mac machines were managed through a paid per-machine console. Wrote a Go agent and browser console for inventory, patching and remote control; it now runs the entire fleet across 17 sites.
- **Cafeteria Management System:** Four cafeterias ran on coupons and registers. Every scan on the Hikvision face readers is now one recorded meal, with live counts, monthly HR reports and roles enforced in Postgres.
- **HikCentral:** The gate access terminal kept event history that could not be searched. Built a dashboard that mirrors it into Postgres for instant search by person, card or event, with live monitoring kept on the device.
- **Sonar:** Orders arrived in dozens of customer WhatsApp groups and were tracked by reading chats. Built a service that captures every order message across groups into one structured, auditable intake feed (Node.js, Baileys).
- **PDFForge:** Staff were uploading confidential documents to public PDF websites. Built a self-hosted platform with 36 tools for merge, split, compress, OCR and conversion, with zero retention, so no document leaves the company.

Founder & Lead Engineer

2024 – Present

3D Cargo Loader Pro, AI logistics SaaS, Jaipur, India

- Built a 3D container-load optimizer with a natural-language interface and live 3D view; plans a load in minutes.
- Finalist, ISF Junicorn Competition, Dubai, selected among top early-stage ventures from dozens of countries.

PROJECTS

ITSM: endpoint fleet management that replaces a commercial RMM

One browser console for every Windows and Mac machine in the company: inventory, patching, live health and remote control. Benchmarked against Kaseya VSA and JP1 ITDM2, and built to go further on agent security.

Stack: Go 1.26 · PostgreSQL 17 · pgx · Next.js 16 · React 19 · Docker (distroless, non-root)

- Single static Go agent for Windows and macOS: outbound-only, signed jobs and updates, tamper-protected.
- One Go binary of six vertical modules (agents, identity, software, remote, security, monitoring); strict CSP throughout.
- Runs 554 machines across 17 sites at 92.4% patch compliance, tested to 5,000 agents; remote shell and USB control.

Aether: meeting, chat and calling platform, an in-house alternative to Microsoft Teams

Complete collaboration software: video meetings with screen share, captions and reactions; persistent chat with channels, groups and files; calling; scheduling; company directory; remote desktop control with consent.

Stack: Go · WebRTC SFU · AV1 / Opus · SRTP · WebSockets · TURN relay · SQLite · Windows desktop app

- A state change costs **3 bytes** on the wire against 172 as JSON, and a heartbeat costs 1 byte, measured on real packets.
- A 4-person, 15-minute call sends **224 MB instead of 365 MB** at the same quality (AV1, 60 ms Opus, pruned headers).
- Directory search over 50,000 people from an index in microseconds; Argon2id, tamper-evident audit log.

Velocity: visual analytics platform that replaces Alteryx Designer

The drag-and-drop canvas analysts already know, rebuilt for the browser and the server. Designed to go beyond Alteryx Designer where teams feel it most: workflows are shared, scheduled and run centrally, with no per-seat licence.

Stack: Python · FastAPI · DuckDB · React · React Flow · scikit-learn · statsmodels · WeasyPrint · Docker

- Compiles the whole canvas into one DuckDB query before reading any data, so a 12-tool pipeline costs one query.
- 157 tools; SQL pushdown to Postgres, MySQL and SQL Server; PDF reports and mail merge; ETS/ARIMA forecasting.

Recon OS: marketplace reconciliation and revenue assurance

Computes what a marketplace was allowed to charge under contract and compares it with what it actually charged, line by line. Where tools such as Sellerboard and Unicommerce report the marketplace's own figures, Recon OS verifies them.

Stack: Python · FastAPI · PostgreSQL 17 · Temporal · MinIO · Next.js · Grafana · 24-module modular monolith

- Verified on a full month of real marketplace files, where it surfaced returns that reached the warehouse but were never credited and settlement rows that did not balance.
- Every figure traces to its source row, match score, contract clause and formula, and the report states what it could not verify instead of reporting complete.
- Multi-tenant with row-level isolation and a hash-chained audit log; a new marketplace is added with a connector spec and a rate card, with no change to the engine.

Additional Software

- **Asset Vault:** Digital asset management for the brand's photography, swatches and product records. Modelled on Pimcore's asset engine and extended with full-text search inside PDF and Office files and per-folder permissions.
- **PriceLens:** Competitive price intelligence that finds the same product across three fashion giants and reports the price gap with a confidence score, so pricing decisions rest on verified matches.
- **Spykar IQ:** Inventory and sales intelligence for a national apparel distribution chain, giving distributors, stores and warehouses one view of stock and a natural-language query engine on Gemini.
- **B2B Dealer Portal:** Order-entry portal for thousands of channel partners that kept orders flowing while the Salesforce B2B portal was under maintenance, with a simpler ordering flow than the portal it stood in for; orders synced back afterwards.
- **Nexus IT** asset lifecycle management · **Voice Flow** system-wide dictation for macOS and Windows · **WhatsApp Marketing Platform**, an in-house replacement for Salesforce Marketing Cloud.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, C, C++, SQL, Bash

AI / ML: PyTorch, TensorFlow, scikit-learn, LightGBM, HuggingFace, PEFT (QLoRA / LoRA), vLLM, LangChain, LangGraph, LlamaIndex, FAISS, RAG, Claude / Gemini / OpenAI APIs

AI Tools: Claude Code, OpenAI Codex, Google Antigravity

Backend: FastAPI, Flask, NestJS, Node.js, Express, Go net/http, REST, GraphQL, WebSockets, WebRTC, Celery, BullMQ, Temporal, JWT, OAuth 2.0, Argon2id · **Frontend:** React, Next.js, Vite, Tailwind CSS, React Flow, ECharts

Data: PostgreSQL, SQL Server, MySQL, MongoDB, Redis, SQLite, DuckDB, Delta Lake, MinIO / S3, Spark, Trino, Airflow, Kafka, Debezium, Superset, Power BI

Infrastructure: Docker, Docker Compose, nginx, GitHub Actions, AWS (EC2, S3), Linux, Windows Services, Wireshark

EDUCATION

B.Tech, Electronics & Communication Engineering, LNMIIT, Jaipur 2024 – 2028